

Strömungslehre I

Formelsammlung

Ein Open-Source-Projekt für Studierende der Universität Stuttgart.

Der Quellcode dieser Formelsammlung steht auf GitHub zur Verfügung. Bei Anmerkungen, Fragen, Kritik und dem Drang, mitzuwirken, kontaktiere gerne den Herausgeber des verlinkten Repositorys.
Für den Studiengang: B.Sc. Luft- und Raumfahrttechnik. Made with L^AT_EX and love.

Terminologie

Begriff	Bedeutung	Mathematisch
Stromlinie	Linie, die zu jedem Zeitpunkt tangential zum örtlichen Geschwindigkeitsvektor verläuft.	$\Psi = \text{konstant}$
Stromfaden	Die Gesamtheit aller Stromlinien, die durch eine Fläche S hindurch treten (besteht aus der Ein- und Austrittsfläche S_1 , S_2 und der Mantelfläche).	-
Stromröhre	Bezeichnung für die röhrenförmige Mantelfläche des Stromfadens.	-
Bahnlinie	Die Wegkurve eines einzelnen, individuellen Masseteilchens.	-
Streichlinie	Der Ort aller Massepunkte in einer Strömung, die an einem bestimmten, festen Punkt vorbei gestrichen sind.	-
Zeitlinie	Linien, welche durch Teilchen- / Blasenbildung gleichzeitig entlang eines gewählten Strömungspotenzials erzeugt werden.	-
Isotache	Linie gleicher Geschwindigkeit in der Strömung.	$ v = \text{konstant}$
Isokline	Linie gleicher Stromlinienneigung in der Strömung.	$\tan(\alpha) = \frac{v}{u} = \text{konst.}$

Symbolverzeichnis

Symbol	SI-Einheit	Bedeutung
c_p	$[-]$	Druckbeiwert
c_p, c_v	$\left[\frac{\text{J}}{\text{kgK}}\right]$	Spezifische, isobare / isochore Wärmekapazität
c_A, c_W	$[-]$	Auftriebs- und Widerstandsbeiwert
F_x, F_y, F_z	$[\text{N}]$	Kraft auf ein Fluid (in x , y oder z -Richtung)
u, v, w	$\left[\frac{\text{m}}{\text{s}}\right]$	Geschwindigkeiten in x , y und z -Richtung
U_∞ / v_∞	$\left[\frac{\text{m}}{\text{s}}\right]$	Geschwindigkeit in x -Richtung, unendlich entfernt
$\dot{\gamma}$	$\left[\frac{1}{\text{s}}\right]$	Schergeschwindigkeit
ζ	$[-]$	Druckverlustbeiwert
η	$[\text{Pa s}]$	Kinematische Viskosität
ϑ	$[-]$	Umlenkwinkel (Bezogen auf Anströmrichtung)
κ	$[-]$	Isentropenexponent
ν	$\left[\frac{\text{m}^2}{\text{s}}\right]$	Dynamische Viskosität (Zähigkeit)
τ / τ_W	$[\text{Pa}]$	Schubspannung / Wandschubspannung
$\underline{x}, \underline{\underline{x}}$		Allgemein: Vektor / Matrix
$\bar{x}$		Allgemein: Gemittelte Größe
$\nabla(\dots)$		Nabla-Operator (Mathematisch)
$\Delta(\dots)$		Gradient (Mathematisch)
$\text{rot}(\dots)$		Rotation (Mathematisch)
$\text{div}(\dots)$		Divergenz (Mathematisch)

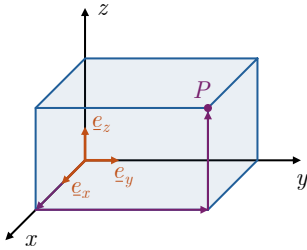
Der Übersicht wegen sind hier nicht alle Symbole aufgeführt.
Es kann zu doppelter Symbolbelegung kommen.

Grundlagen

Koordinatensysteme

Kartesische Koordinaten

$$\begin{aligned}v_x &= u \\v_y &= v \\v_z &= w\end{aligned}$$



$$h_1 = 1 \quad h_2 = 1 \quad h_3 = 1$$

$$\nabla = \underline{e}_x \frac{\partial}{\partial x} + \underline{e}_y \frac{\partial}{\partial y} + \underline{e}_z \frac{\partial}{\partial z}$$

$$\nabla \cdot \underline{v} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$$

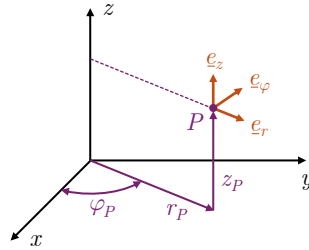
$$(\nabla \times \underline{v})_x = \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}$$

$$(\nabla \times \underline{v})_y = \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}$$

$$(\nabla \times \underline{v})_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Zylinderkoordinaten

$$\begin{aligned}x &= r \cos(\varphi) & r &= \sqrt{x^2 + y^2} \\y &= r \sin(\varphi) & \varphi &= \arctan\left(\frac{y}{x}\right) \\z &= z & z &= z\end{aligned}$$



$$h_1 = 1 \quad h_2 = r \quad h_3 = 1$$

$$\nabla = \underline{e}_r \frac{\partial}{\partial r} + \underline{e}_\varphi \frac{1}{r} \frac{\partial}{\partial \varphi} + \underline{e}_z \frac{\partial}{\partial z}$$

$$\nabla \cdot \underline{v} = \frac{1}{r} \frac{\partial}{\partial r} (r u) + \frac{1}{r} \frac{\partial v}{\partial \varphi} + \frac{\partial w}{\partial z}$$

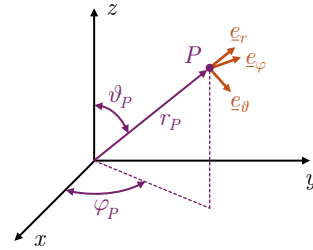
$$(\nabla \times \underline{v})_r = \frac{1}{r} \frac{\partial w}{\partial \varphi} - \frac{\partial v}{\partial z}$$

$$(\nabla \times \underline{v})_\varphi = \frac{\partial u}{\partial z} - \frac{\partial w}{\partial r}$$

$$(\nabla \times \underline{v})_z = \frac{1}{r} \left[\frac{\partial}{\partial r} (r u) - \frac{\partial u}{\partial \varphi} \right]$$

Kugelkoordinaten

$$\begin{aligned}x &= r \sin(\vartheta) \cos(\varphi) & r &= \sqrt{x^2 + y^2 + z^2} \\y &= r \sin(\vartheta) \sin(\varphi) & \vartheta &= \arccos\left(\frac{z}{r}\right) \\z &= r \cos(\varphi) & \varphi &= \arccos\left(\frac{x}{\sqrt{x^2 + y^2}}\right)\end{aligned}$$



$$h_1 = 1 \quad h_2 = r \quad h_3 = r \sin(\vartheta)$$

$$\nabla = \underline{e}_r \frac{\partial}{\partial r} + \underline{e}_\vartheta \frac{1}{r} \frac{\partial}{\partial \vartheta} + \underline{e}_\varphi \frac{1}{r \sin(\vartheta)} \frac{\partial}{\partial \varphi}$$

$$\nabla \cdot \underline{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 u) + \frac{1}{r \sin(\vartheta)} \frac{\partial}{\partial \vartheta} (v \sin(\vartheta)) + \frac{1}{r \sin(\vartheta)} \frac{\partial w}{\partial \varphi}$$

$$(\nabla \times \underline{v})_r = \frac{1}{r \sin(\vartheta)} \left[\frac{\partial}{\partial \vartheta} (\sin(\vartheta) w) - \frac{\partial v}{\partial \varphi} \right]$$

$$(\nabla \times \underline{v})_\vartheta = \frac{1}{r \sin(\vartheta)} \frac{\partial u}{\partial \varphi} - \frac{1}{r} \frac{\partial}{\partial r} (r w)$$

$$(\nabla \times \underline{v})_\varphi = \frac{1}{r} \left[\frac{\partial}{\partial r} (r v) - \frac{\partial u}{\partial \vartheta} \right]$$

Skalarprodukt

$$\underline{a} \cdot \underline{b}$$

Kreuzprodukt

$$\underline{a} \times \underline{b}$$

Dyadisches Produkt

$$\underline{a} \underline{b} = \underline{a} \otimes \underline{b} = \underline{a} \underline{b}^T$$

Nabla-Operator

$$\nabla = \sum_{i=1}^3 \underline{e}_i \frac{1}{h_i} \frac{\partial}{\partial x_i}$$

Gradient eines Skalars

$$\text{grad}(f) = \nabla(f) = \underline{e}_1 \frac{1}{h_1} \frac{\partial f}{\partial x_1} + \underline{e}_2 \frac{1}{h_2} \frac{\partial f}{\partial x_2} + \underline{e}_3 \frac{1}{h_3} \frac{\partial f}{\partial x_3}$$

Gradient eines Vektors

$$\text{grad}(\underline{v}) = \nabla \otimes \underline{v} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} & \frac{\partial w}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} & \frac{\partial w}{\partial y} \\ \frac{\partial u}{\partial z} & \frac{\partial v}{\partial z} & \frac{\partial w}{\partial z} \end{pmatrix}$$

Divergenz

$$\text{div}(\underline{a}) = \nabla \cdot \underline{a} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial x_1} (h_2 h_3 a_1) + \frac{\partial}{\partial x_2} (h_3 h_1 a_2) + \frac{\partial}{\partial x_3} (h_1 h_2 a_3) \right]$$

Rotation

$$\text{rot}(\underline{a}) = \nabla \times \underline{a} = \frac{1}{h_1 h_2 h_3} \left[h_1 \left(\frac{\partial (h_3 a_3)}{\partial x_2} - \frac{\partial (h_2 a_2)}{\partial x_3} \right) \underline{e}_1 + h_2 \left(\frac{\partial (h_1 a_1)}{\partial x_3} - \frac{\partial (h_3 a_3)}{\partial x_1} \right) \underline{e}_2 + h_3 \left(\frac{\partial (h_2 a_2)}{\partial x_1} - \frac{\partial (h_1 a_1)}{\partial x_2} \right) \underline{e}_3 \right]$$

Laplace-Operator

$$\nabla^2(f) = \Delta(f) = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial x_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial x_1} \right) + \frac{\partial}{\partial x_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial x_2} \right) + \frac{\partial}{\partial x_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial x_3} \right) \right]$$

Physikalische Eigenschaften

Knudsen-Zahl

$$\text{Kn} = \frac{\bar{\lambda}}{L}$$

$\bar{\lambda}$ = Mittlere freie Weglänge
 L = Charakteristische Länge

Kontinuumsströmung: $\text{Kn} < 10^{-2}$
 Freie Molekülströmung: $\text{Kn} > 10$

Höhe	0 km	80 km	150 km
$\bar{\lambda}$	3 nm	100 nm	2000 nm
L	1,6 μm	3 cm	200 m

Dichte

Flüssigkeiten: $\rho = \text{konst.}$
 Barotrope Fluide: $\rho = \rho(p)$
 Gase: $\rho = \rho(p, T)$

Druck

$$p_{\text{ges}} = p_{\text{stat}} + p_{\text{dyn}}$$

$$p_{\text{dyn}} = q = \frac{1}{2} \rho v^2$$

Ideale Gasgleichung

$$p = \rho R T$$

$$p V = m R T$$

Spezifische Gaskonstante

$$R = \frac{\Re}{M} = c_p - c_v$$

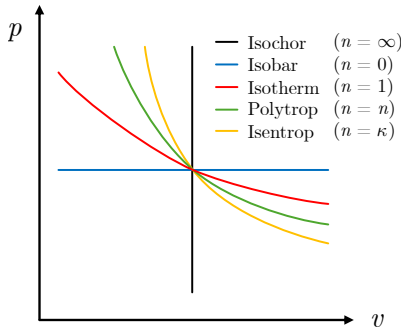
$$\Re = 8314,3 \frac{\text{J}}{\text{kmol K}}$$

M = Molare Masse

c_p = Spezifische Wärme bei konstantem Druck

c_v = Spezifische Wärme bei konstanter Temperatur

Polytrope Zustandsänderung



$$\frac{p}{\rho^n} = n v^n = \text{konst.}$$

$$\frac{\rho}{\rho_b} = \left(\frac{p}{p_b} \right)^{\frac{1}{n}}$$

n = Polytropenexponent

κ = Isentropenexponent

$$\kappa = \frac{c_p}{c_v} = \frac{c_p}{c_p - R} = \frac{c_v + R}{c_v}$$

Schallgeschwindigkeit

$$c = \sqrt{\left(\frac{\partial p}{\partial \rho} \right)_s} = \sqrt{\kappa \left(\frac{\partial p}{\partial \rho} \right)_T}$$

$$= \underbrace{\sqrt{\kappa R T}}_{\text{Ideales Gas}} = \sqrt{\kappa \frac{p}{\rho}}$$

$$\frac{\Delta \rho}{\rho} \approx \frac{\text{Ma}^2}{2}$$

Grenze Inkompressibler Strömung

Temperatur

$$E_{\text{kin}} = \frac{3}{2} k_B T$$

E_{kin} = kinetische Energie

k_B = Boltzmann-Konstante

$$= 1,38 \cdot 10^{-23} \frac{\text{J}}{\text{K}}$$

T = Temperatur

Newton'sches Reibungsgesetz

$$\tau = \eta \frac{du}{dy} = \eta D$$

τ = Schubspannung

η = Dynamische Viskosität $\left[\frac{\text{kg}}{\text{m s}} \right]$

D = Schergeschwindigkeit

Kinematische Viskosität

$$\nu = \frac{\eta}{\rho}$$

ν = Kinematische Viskosität $\left[\frac{\text{m}^2}{\text{s}} \right]$

η = Dynamische Viskosität $\left[\frac{\text{kg}}{\text{m s}} \right]$

ρ = Dichte

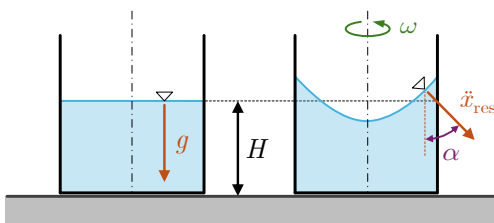
Fluidstatik & Hydrostatik

Grenzfläche (Rotierender Zylinder)

$$F_z = m \omega^2 r = m \frac{v^2}{r} \quad \left| \quad z(r) = H + \frac{\omega^2}{2g} \left(r^2 - \frac{1}{2} R^2 \right) \right.$$

$$F_G = m g$$

$$\tan(\alpha) = \frac{F_z}{F_G} = \frac{\omega^2 r}{g}$$



Flüssigkeitssäule

$$F_x = \frac{\rho g}{2} (y_2^2 - y_1^2) dz$$

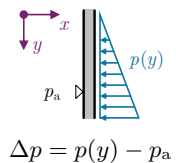
$$dF_x = \Delta p dA_y$$

$$F_y = \rho g dz \int_{x_1}^{x_2} y_W(x) dx$$

$$dF_y = -\Delta p dA_x$$

$$M_A = \rho g dz \left[\int_{x_1}^{x_2} x y_W(x) dx - \frac{1}{3} (y_2^3 - y_1^3) \right]$$

$$dM_A = \begin{pmatrix} x \\ y \end{pmatrix} \times \begin{pmatrix} dF_x \\ dF_y \end{pmatrix} = x dF_y - y dF_x$$



Hydrostatischer Auftrieb

Sinken / Schweben

$$F_A = (\rho_F - \rho_K) g V_v$$

F_A = Auftriebskraft

ρ_F = Dichte (Fluid)

ρ_K = Dichte (Körper)

Schwimmen

$$\rho_F g V_v = \rho_K g V_K$$

g = Erdbeschleunigung

V_v = Verdrängtes Volumen

V_K = Volumen (Körper)

Atmosphäre

Isotherm

$$p(z) = p_0 \exp\left(-\frac{\rho_0 g}{p_0} z\right)$$

$$\rho(z) = \rho_0 \exp\left(-\frac{\rho_0 g}{p_0} z\right)$$

$$T(z) = T_0$$

Isentrop

$$p(z) = p_0 \left(1 - \frac{\kappa - 1}{\kappa} \frac{\rho_0 g}{p_0} z \right)^{\frac{\kappa}{\kappa - 1}}$$

$$\rho(z) = \rho_0 \left(1 - \frac{\kappa - 1}{\kappa} \frac{\rho_0 g}{p_0} z \right)^{\frac{\kappa}{\kappa - 1}}$$

$$T(z) = T_0 \left(1 - \frac{\kappa - 1}{\kappa} \frac{\rho_0 g}{p_0} z \right)$$

Kontinuitätsgleichung

Integralform (Konservativ)

$$\frac{\partial}{\partial t} \iiint_V \rho \, dV + \oint_S \rho \underline{v} \cdot d\underline{S} = 0$$

Stromröhre

$$\dot{m} = \rho v S = \text{konst.}$$

Stationär

$$\text{div}(\rho \underline{v}) = 0$$

Inkompressibel

$$\text{div}(\underline{v}) = 0$$

Differentialform

Konservativ

$$\frac{\partial \rho}{\partial t} + \text{div}(\rho \underline{v}) = \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \underline{v}) = 0$$

Nicht-Konservativ

$$\frac{D\rho}{Dt} + \rho \text{div}(\underline{v}) = \frac{D\rho}{Dt} + \rho \nabla \cdot \underline{v} = 0$$

Zylinderkoordinaten

$$\frac{\partial \rho}{\partial t} + \left[\frac{1}{r} \frac{\partial}{\partial r} (r \rho u) + \frac{1}{r} \frac{\partial}{\partial \varphi} (\rho v) + \frac{\partial}{\partial z} (\rho w) \right]$$

Kugelkoordinaten

$$\frac{\partial \rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \rho u) + \frac{1}{r \sin(\vartheta)} \frac{\partial}{\partial \vartheta} (\rho v \sin(\vartheta)) + \frac{1}{r \sin(\vartheta)} \frac{\partial}{\partial \varphi} (\rho w) = 0$$

Impulssatz

Differentialform

Konservativ

$$\begin{aligned} \frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \underline{v}) &= -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho f_x \\ \frac{\partial(\rho v)}{\partial t} + \nabla \cdot (\rho v \underline{v}) &= -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + \rho f_y \\ \frac{\partial(\rho w)}{\partial t} + \nabla \cdot (\rho w \underline{v}) &= -\frac{\partial p}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} + \rho f_z \end{aligned}$$

Nicht-Konservativ

$$\rho \frac{D\underline{v}}{Dt} = \text{div}(\underline{\sigma}) + \rho \underline{f}$$

$$\begin{aligned} \rho \frac{D(\rho u)}{Dt} &= -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho f_x \\ \rho \frac{D(\rho v)}{Dt} &= -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + \rho f_y \\ \rho \frac{D(\rho w)}{Dt} &= -\frac{\partial p}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} + \rho f_z \end{aligned}$$

Integralform

$$\iiint_V \frac{\partial(\rho \underline{v})}{\partial t} \, dV + \oint_S \rho \underline{v} (\underline{v} \cdot \underline{n}) \, dS = \iiint_V \rho \underline{f} \, dV + \oint_S \underline{\sigma} \cdot d\underline{S}$$

Stationär

$$\frac{d\underline{I}}{dt} = \oint_S \rho \underline{v} (\underline{v} \cdot \underline{n}) \, dS$$

Reynold'sches Transporttheorem

$$\frac{DB}{Dt} = \frac{D}{Dt} \iiint_{V(t)} b(t) \, dV = \iiint_{V_0} \frac{\partial b}{\partial t} \, dV + \iiint_{V_0} \text{div}(b \underline{v}) \, dV$$

$$\frac{D\underline{B}}{Dt} = \frac{D}{Dt} \iiint_{V(t)} \underline{b}(t) \, dV = \iiint_{V_0} \frac{\partial \underline{b}}{\partial t} \, dV + \oint_S \underline{b} (\underline{v} \cdot \underline{n}) \, dS$$

Praktischer Einsatz

Wahl des Koordinatensystems

Wahl des Kontrollvolumens (KV)

Zusammenstellung der Kräfte
 $\underline{F} = \underline{F}_p + \underline{F}_f + \underline{F}_K + \underline{F}_R$

Lösung der Gleichung

$$\iiint_V \frac{\partial(\rho \underline{v})}{\partial t} \, dV + \oint_S \rho \underline{v} (\underline{v} \cdot \underline{n}) \, dS = \underline{F}$$

Druckkräfte F_p

$$d\underline{F}_p = -p \cdot \underline{n} \, dS$$

$$\underline{F}_p = - \oint_S p \underline{n} \, dS$$

Volumenkräfte F_f

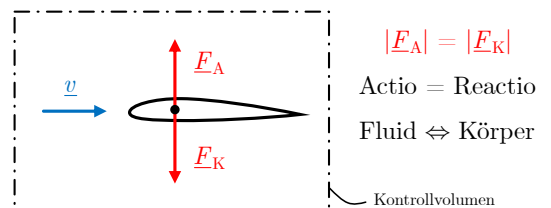
$$\underline{F}_f = \iiint_V \rho \underline{f} \, dV$$

Schwerefeld

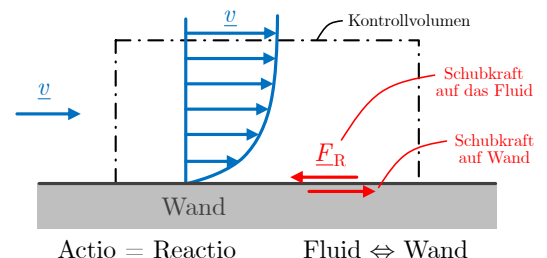
$$d\underline{F}_f = \rho \underline{g} \, dV$$

$$\underline{F}_f = \iiint_V \rho \underline{g} \, dV$$

Körperkräfte F_K



Reibungskräfte F_R



Carnot-Diffusor

Druckdifferenz

$$\Delta p_v = \frac{\rho}{2} u_1^2 \left(1 - \frac{u_2}{u_1} \right)^2$$

Druckverlustbeiwert

$$\zeta = \frac{2 \Delta p_v}{\rho u_1^2} = \left(1 - \frac{A_1}{A_2} \right)^2$$

Impulssatz

Eulergleichungen

Konservativ	Nicht-Konservativ
$\frac{\partial(\rho \underline{v})}{\partial t} + \operatorname{div}(\rho \cdot \underline{v} \otimes \underline{v}) = -\nabla(p) + \rho \underline{f}$	$\rho \frac{D\underline{v}}{Dt} = \rho \left[\frac{\partial \underline{v}}{\partial t} + \underline{v} \cdot \nabla(\underline{v}) \right] = -\nabla(p) + \rho \underline{f}$
$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\partial(\rho u v)}{\partial y} + \frac{\partial(\rho u w)}{\partial z} = -\frac{\partial p}{\partial x} + \rho f_x$	$\rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} + \rho w \frac{\partial u}{\partial z} = -\frac{\partial p}{\partial x} + \rho f_x$
$\frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho v u)}{\partial x} + \frac{\partial(\rho v^2)}{\partial y} + \frac{\partial(\rho v w)}{\partial z} = -\frac{\partial p}{\partial y} + \rho f_y$	$\rho \frac{\partial v}{\partial t} + \rho u \frac{\partial v}{\partial x} + \rho v \frac{\partial v}{\partial y} + \rho w \frac{\partial v}{\partial z} = -\frac{\partial p}{\partial y} + \rho f_y$
$\frac{\partial(\rho w)}{\partial t} + \frac{\partial(\rho w u)}{\partial x} + \frac{\partial(\rho w v)}{\partial y} + \frac{\partial(\rho w^2)}{\partial z} = -\frac{\partial p}{\partial z} + \rho f_z$	$\rho \frac{\partial w}{\partial t} + \rho u \frac{\partial w}{\partial x} + \rho v \frac{\partial w}{\partial y} + \rho w \frac{\partial w}{\partial z} = -\frac{\partial p}{\partial z} + \rho f_z$

Bernoulligleichungen

Stationär, Inkompressibel, Reibungsfrei

Allgemein	Energieform	Druckform	Höhenform
$\rho_0 \frac{v^2}{2} + p + \rho_0 U = \text{konst.}$	$\frac{v^2}{2} + \frac{p}{\rho} + g z = \text{konst.}$	$\frac{\rho}{2} v^2 + p + \rho g z = \text{konst.}$	$\frac{v^2}{2g} + \frac{p}{\rho g} + z = \text{konst.}$

Instationär, Inkompressibel, Reibungsfrei		Instationär, Kompressibel, Reibungsfrei
Allgemein	Querdruckgleichung	Allgemein
$\int_1^2 \frac{\partial v}{\partial t} ds + \frac{v_2^2 - v_1^2}{2} + \frac{p_2 - p_1}{\rho} + g(z_2 - z_1) = 0$	$\frac{\partial v_n}{\partial t} - \frac{v^2}{r} + \frac{1}{\rho} \frac{\partial p}{\partial n} + g \frac{\partial z}{\partial n} = 0$	$\int_1^2 \frac{\partial v}{\partial t} ds + \frac{v_2^2 - v_1^2}{2} + \int_1^2 \frac{dp}{\rho} + U_2 - U_1 = 0$

Anwendungen

Venturi-Rohr		Ausströmen aus Gefäß	
Druckdifferenz	Massenstrom	Konstanter Spiegel	Mit Spiegelabsenkung
$\Delta p = \frac{\rho}{2} v_1^2 \left[1 - \left(\frac{A_1}{A_2} \right)^2 \right]$	$\dot{m} = A_1 A_2 \sqrt{\frac{2 \rho \Delta p}{A_2^2 - A_1^2}}$	$v_2 = \sqrt{\frac{2}{\rho} (p_1 - p_2) + 2 g h}$	$\Delta t = \frac{A_1}{A_2} \sqrt{\frac{2}{g}} (\sqrt{z_1} - \sqrt{z_2})$

Prandtl-Rohr		Schwingende Säule
Inkompressibel		$x = x_0 \cos(\omega t)$
Druckdifferenz	Anströmgeschwindigkeit	
$\Delta p = p_0 - p_\infty = \rho_F g \Delta z$	$v_\infty = \sqrt{\frac{2 \Delta p}{\rho}} = \sqrt{2 \frac{\rho_F}{\rho} g \Delta z}$	

Kompressibel		Bewegung auf konzentrischen Bahnen
Fehlerabschätzung	Anströmgeschwindigkeit	$p(r) = p_1 + \frac{\rho}{2} v_1^2 \left(1 - \frac{r_1^2}{r^2} \right)$
$\frac{v_{\infty, \text{komp.}}}{v_{\infty, \text{inkomp.}}} \approx \sqrt{1 - \frac{\text{Ma}_\infty^2}{4}}$	$v_\infty = \sqrt{\frac{2 \kappa}{\kappa - 1} \frac{p_\infty}{\rho_\infty} \left[\left(\frac{p_0}{p_\infty} \right)^{\frac{\kappa - 1}{\kappa}} - 1 \right]}$	

	Symbol	Inkompressibel	Kompressibel (Ma ≤ 1)	Kompressibel (Ma > 1)
Statischer Druck	p, p_s, p_{stat}	p	p	p
Dynamischer Druck	q, p_{dyn}	$p_{\text{dyn}} = p_0 - p = \frac{\rho}{2} v^2$	$p_{\text{dyn}} = p_0 - p > \frac{\rho}{2} v^2$	
Totaldruck	p_0, p_t, p_{ges}	$p_0 = p + \frac{\rho}{2} v^2$	$p_0 = p + \frac{\rho}{2} v^2 + \Delta p_{\text{komp.}} = p \left(1 + \frac{\kappa - 1}{2} \text{Ma}^2 \right)^{\frac{\kappa - 1}{\kappa}}$	
Pitotdruck	p_{pitot}	$p_{\text{pitot}} = p_0$	$p_{\text{pitot}} = p_0$	$p_{\text{pitot}} \leq p_0$

Navier-Stokes-Gleichungen

Scherraten-Tensor	Drehgeschwindigkeits-Tensor
$\underline{\underline{D}} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{\partial v}{\partial y} & \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) & \frac{\partial w}{\partial z} \end{pmatrix}$	$\underline{\underline{W}} = \begin{pmatrix} 0 & \frac{1}{2} \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) & 0 & \frac{1}{2} \left(\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \right) \\ \frac{1}{2} \left(\frac{\partial w}{\partial x} - \frac{\partial u}{\partial z} \right) & \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) & 0 \end{pmatrix}$
Spannungs-Tensor	
$\underline{\underline{\sigma}} = \begin{pmatrix} -p + 2\eta \left(\frac{\partial u}{\partial x} - \frac{1}{3} \text{div}(\underline{v}) \right) & \eta \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \eta \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \\ \eta \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) & -p + 2\eta \left(\frac{\partial v}{\partial y} - \frac{1}{3} \text{div}(\underline{v}) \right) & \eta \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \\ \eta \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) & \eta \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right) & -p + 2\eta \left(\frac{\partial w}{\partial z} - \frac{1}{3} \text{div}(\underline{v}) \right) \end{pmatrix}$	
Divergenz des Spannungstensors	Inkompressible Strömung
$\text{div}(\underline{\underline{\sigma}}) = -\nabla(p) + 2 \text{div}(\eta \underline{\underline{D}}) - \frac{2}{3} \nabla(\eta \text{div}(\underline{v}))$	$\frac{\partial \underline{v}}{\partial t} + \underline{v} \cdot \nabla(\underline{v}) = -\nabla \left(\frac{p}{\rho} \right) + \nu \Delta(\underline{v}) + \underline{f}$

Kartesische Koordinaten (Newton'sche Fluide)

$$\begin{aligned} \rho \left[\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right] &= -\frac{\partial p}{\partial x} + \rho f_x + \frac{\partial}{\partial x} \left[2\eta \left(\frac{\partial u}{\partial x} - \frac{\text{div}(\underline{v})}{3} \right) \right] + \frac{\partial}{\partial y} \left[\eta \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \right] + \frac{\partial}{\partial z} \left[\eta \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \right] \\ \rho \left[\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right] &= -\frac{\partial p}{\partial y} + \rho f_y + \frac{\partial}{\partial x} \left[\eta \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \right] + \frac{\partial}{\partial y} \left[2\eta \left(\frac{\partial v}{\partial y} - \frac{\text{div}(\underline{v})}{3} \right) \right] + \frac{\partial}{\partial z} \left[\eta \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \right] \\ \rho \left[\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right] &= -\frac{\partial p}{\partial z} + \rho f_z + \frac{\partial}{\partial x} \left[\eta \left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} \right) \right] + \frac{\partial}{\partial y} \left[\eta \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right) \right] + \frac{\partial}{\partial z} \left[2\eta \left(\frac{\partial w}{\partial z} - \frac{\text{div}(\underline{v})}{3} \right) \right] \end{aligned}$$

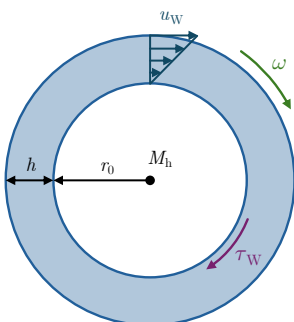
Kartesische Koordinaten (Inkompressible Fluide)

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + f_x \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + f_y \\ \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} &= -\frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + f_z \end{aligned}$$

Couette-Viskosimeter

$$\begin{aligned} u &= U \frac{y}{h} \\ \tau_W &= \frac{M_h}{2\pi r_0^2 b} \quad \left| \quad \eta = \frac{\tau_W}{\dot{\gamma}} \right. \\ \dot{\gamma} &= \frac{u_W}{h} \quad \left| \quad \nu = \frac{\eta}{\rho} = \frac{\tau_W}{\rho \dot{\gamma}} \right. \end{aligned}$$

M_h = Haltemoment
 $\dot{\gamma}$ = Schergeschwindigkeit
 η = Dynamische Viskosität
 ν = Kinematische Viskosität



Gesetz von Hagen-Poiseuille

Radialgeschwindigkeit	Länge des Übergangsbereiches
$w = \frac{a}{4} (r^2 - R^2) = w_{\max} \left(1 - \frac{r^2}{R^2} \right)$	$L_{12} \approx a D \left(\frac{\bar{w} D}{\nu} \right)^b = a D \text{Re}_D^b$
Massenstrom	Reynoldszahl
$\dot{m} = \frac{1}{2} \rho \pi w_{\max} R^2 = \rho \pi R^2 \bar{w}$	$\text{Re}_D = \frac{\rho \bar{w} D}{\eta} = \frac{\bar{w} D}{\nu}$
Druckdifferenz	Nach Dürst
$\Delta p = -\frac{4\eta w_{\max}}{R^2} L = -\frac{8\eta \dot{m} L}{\rho \pi R^4} = -\frac{64}{\underbrace{\text{Re}_D}_{\lambda}} \frac{\rho L}{2D} \bar{w}^2$	$L_{12} = D \sqrt[3]{C_0^\beta + (C_1 \text{Re}_D)^\beta}$
Allgemein	Rohrströmung
$\eta \frac{\partial}{\partial r} \left(r \frac{\partial w}{\partial r} \right) = \frac{\partial p}{\partial z}$	$\beta = 1,6$ $C_0 = 0,619$ $C_1 = 0,0567$
Wandschubspannung	Kanalströmung
$\tau_W = -\frac{R}{2} \frac{\Delta p}{L}$	$\beta = 1,6$ $C_0 = 0,631$ $C_1 = 0,0442$
Einlaufströmung:	Nach Kuhlmann
$p_1 = p_0 - \frac{\rho}{2} \bar{w}^2$	$L_{12} \approx \frac{\text{Re}_D}{50} D$

Ähnlichkeitstheorie

Reynoldszahl	Strouhal-Zahl	Machzahl	Froude-Zahl	Stokes'sche Formel
$Re_L = \frac{\rho v_\infty L}{\eta} = \frac{v_\infty L}{\nu}$	$St = \frac{L f}{v_\infty}$	$Ma = \frac{v_\infty}{c} = \frac{v_\infty}{\sqrt{\kappa R T}}$	$Fr = \frac{v_\infty}{\sqrt{g L}}$	$F_W = \frac{\rho}{2} v_\infty^2 c_W S = 6 \pi \eta R v_\infty$

Vorgehensweise bei Modellversuchen



Grenzschichten

Wandschubspannung	Grenzschichtdicke	Widerstandsbeiwert
$\tau_0 = \frac{\rho v_\infty^2}{6} \frac{d\delta}{dx}$	$\frac{\delta}{x} = \sqrt{\frac{12 \nu}{v_\infty x}} = \sqrt{\frac{12}{Re_x}}$	Lokal $c_f = \frac{2}{\sqrt{12 Re_x}} = \frac{2 \tau_0(x)}{\rho v_\infty^2}$ Gesamt $c_W = 2 \sqrt{\frac{\nu}{3 v_\infty x}} = \frac{2}{\sqrt{3 Re_x}}$

Grenzschichtgleichungen

Kontinuitätsgleichung	y-Impuls	x-Impuls	Randbedingungen
$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$	$\frac{\partial p}{\partial y} = 0$	$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \frac{\partial^2 u}{\partial y^2}$	$\underbrace{u = v = 0}_{y=0}$ $\underbrace{u = u_\delta(x)}_{y=\delta}$

Blasius-Gleichung

Variable	Stromfunktion	Blasius-Gleichung	Randbedingungen
$\eta = y \sqrt{\frac{v_\infty}{\nu x}} = \frac{y}{\delta}$	$\Psi(x, y) = \sqrt{v_\infty \nu x} \cdot f(\eta)$	$f''' + \frac{1}{2} f f'' = 0$	$y=0 / \eta=0: \quad u=v=f(0)=f'(0)=0$ $y \rightarrow \infty / \eta \rightarrow \infty: \quad u=v_\infty \mid f'(\eta \ll 1) = 1$

Lösungen

η	f	f'	f''
0	0	0	0,3321
0,5	0,0415	0,1659	0,3309
1,0	0,1656	0,3298	0,3230
1,5	0,3701	0,4868	0,3026
2,0	0,6500	0,6298	0,2668
2,5	0,9963	0,7513	0,2174
3,0	1,3968	0,8460	0,1614
3,5	1,8377	0,9130	0,1078
4,0	2,3057	0,9555	0,0642
4,5	2,7901	0,9795	0,0340
5,0	3,2833	0,9915	0,0159
5,5	3,7806	0,9969	0,0066
6,0	4,2796	0,9990	0,0024
6,5	4,7793	0,9997	0,0008
7,0	5,2792	0,9999	0,0002
7,5	5,7792	1,0000	0,0001
8,0	6,2792	1,0000	0,0000

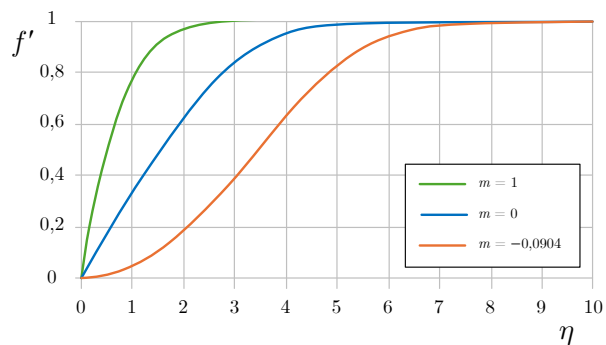
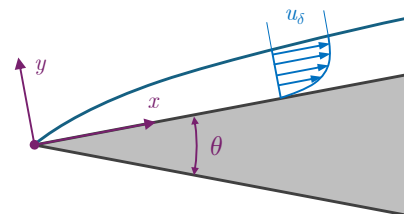
Ebene Platte (Laminar)

Wandschubspannung	Lokaler Widerstandsbeiwert	Grenzschichtdicke
$\tau_0 = \frac{0,3321 \rho v_\infty^2}{\sqrt{Re_x}}$	$c_f = \frac{0,6642}{\sqrt{Re_x}}$ $c_W = \frac{1,328}{\sqrt{Re_x}}$	$\delta = \frac{5x}{\sqrt{Re_x}} \sim x^{0,5}$

Keil

$$m = \frac{\theta}{2\pi - \theta}$$

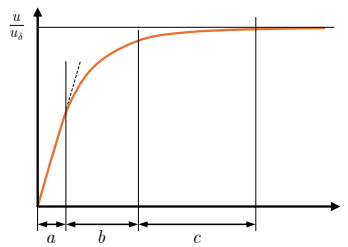
$$u_\delta \approx x^m$$



$m = 1$: Ebene Staupunktströmung
 $m = 0$: Ebene Platte

$m = -0,0904$: Grenzfall
 $m < -0,0904$: Strömungsablösung

Turbulenz

<u>Reynoldszahl</u> $Re_x = \frac{v_\infty x}{\nu}$		<u>Turbulenzgrad (Anströmung)</u> $Tu = \frac{1}{\sqrt{3} U_\infty} \sqrt{\overline{u'^2} + \overline{v'^2} + \overline{w'^2}}$		<u>e^N-Methode</u> $e^N = \exp\left(\int_{t_{ind}}^{t_{krit}} \beta_i dt\right) = \frac{A(x_{krit})}{A(x_{ind})}$											
<u>Re_{x,krit}</u> Ebene Platte : 3,5 · 10 ⁵ - 3,2 · 10 ⁶ Rohr : 2,3 · 10 ³ - 4 · 10 ⁴		<u>Schubspannungsgeschwindigkeit</u> $u_\tau = \sqrt{\frac{\tau_0}{\rho}}$													
<u>Reynolds-gemittelte Navier-Stokes-Gleichungen</u> $\rho \bar{u} \frac{\partial \bar{u}}{\partial x} + \rho \bar{v} \frac{\partial \bar{u}}{\partial y} = -\frac{\partial \bar{p}}{\partial x} + \eta \left[\frac{\partial^2 \bar{u}}{\partial x^2} + \frac{\partial^2 \bar{u}}{\partial y^2} \right] - \rho \frac{\partial}{\partial x} (\overline{u'^2}) - \rho \frac{\partial}{\partial y} (\overline{u' v'})$ $\rho \bar{u} \frac{\partial \bar{v}}{\partial x} + \rho \bar{v} \frac{\partial \bar{v}}{\partial y} = -\frac{\partial \bar{p}}{\partial y} + \eta \left[\frac{\partial^2 \bar{v}}{\partial x^2} + \frac{\partial^2 \bar{v}}{\partial y^2} \right] - \rho \frac{\partial}{\partial x} (\overline{u' v'}) - \rho \frac{\partial}{\partial y} (\overline{v'^2})$			<u>Turbulenter Spannungstensor</u> $\tau_{turb} = -\rho \begin{pmatrix} \overline{u'^2} & \overline{u' v'} & \overline{u' w'} \\ \overline{u' v'} & \overline{v'^2} & \overline{v' w'} \\ \overline{u' w'} & \overline{v' w'} & \overline{w'^2} \end{pmatrix}$												
<u>Reynoldsspannungen</u> $\tau_{xx, turb} = -\rho \overline{u'^2}$ $\tau_{yx, turb} = -\rho \overline{u' v'}$		<u>Ansatz von Boussinesq</u> $\tau_{ges} = \tau_{yx} + \tau_{yx, turb} = (\eta + \eta') \frac{\partial \bar{u}}{\partial y}$		<u>Scherrate</u> $\frac{\partial \bar{u}}{\partial y} = -\frac{\rho \overline{u' v'}}{\eta'}$											
<u>Universelles Wandgesetz</u> $u = \frac{\tau_0 y}{\eta} = \frac{\tau_0 y}{\rho \nu} = \frac{u_\tau^2 y}{\nu}$ $\Rightarrow \underbrace{\frac{u}{u_\tau}}_{u^+} = \underbrace{\frac{u_\tau y}{\nu}}_{y^+}$ $u^+ = \text{Geschwindigkeitsprofil}$ $y^+ = \text{Ausdehnung von Gebiet a}$		<u>Prandtl'sche Mischungsweghypothese</u> $\overline{u' v'} = -l^2 \left(\frac{\partial u}{\partial y} \right) \left \frac{\partial u}{\partial y} \right $ <u>Prandtl'sche Mischungsweglänge</u> $u' = -l \frac{\partial u}{\partial y} = k y \frac{\partial u}{\partial y} \approx 0,41 y \frac{\partial u}{\partial y}$		<u>Logarithmisches Wandgesetz</u> $\frac{u}{u_\tau} = \frac{\ln(10)}{k} \log\left(\frac{u_\tau y}{\nu}\right) + C_2$ $\approx 5,75 \log\left(\frac{u_\tau y}{\nu}\right) + 5,2$ <u>Lokaler Widerstandsbeiwert</u> $c_f = \frac{0,370}{\log(Re_x)^{2,58}} \quad \left \quad c_w = \frac{0,455}{\log(Re_L)^{2,58}} \right.$											
<u>Gebiete</u>  a : Laminare Unterschicht (UWG) b : Wandnahes Gebiet (LWG) c : Außenbereich (PTG)			<u>Potenzgesetz</u> <u>Rohr</u> $\frac{u}{u_{max}} = \left(\frac{y}{R}\right)^{\frac{1}{n}}$ <table><tr><td>Re</td><td>4 · 10³</td><td>1 · 10⁵</td><td>2 · 10⁶</td></tr><tr><td>n</td><td>6</td><td>7</td><td>10</td></tr></table> <u>Platte</u> $\frac{u}{u_\infty} = \left(\frac{y}{\delta}\right)^{\frac{1}{n}} \quad n \approx 7 - 8$ <table><tr><td><u>Grenzschichtdicke</u></td></tr><tr><td>$\delta = \frac{0,37 x}{\sqrt[5]{Re_x}} \sim x^{0,8}$</td></tr></table> <u>Lokaler Widerstandsbeiwert</u> $c_f = \frac{0,0576}{\sqrt[5]{Re_x}} \quad \left \quad c_w = \frac{0,074}{\sqrt[5]{Re_L}} \right.$			Re	4 · 10 ³	1 · 10 ⁵	2 · 10 ⁶	n	6	7	10	<u>Grenzschichtdicke</u>	$\delta = \frac{0,37 x}{\sqrt[5]{Re_x}} \sim x^{0,8}$
Re	4 · 10 ³	1 · 10 ⁵	2 · 10 ⁶												
n	6	7	10												
<u>Grenzschichtdicke</u>															
$\delta = \frac{0,37 x}{\sqrt[5]{Re_x}} \sim x^{0,8}$															
<u>Wandschubspannung</u> <u>Platte</u> $\tau_0 = 0,0225 \rho u_\infty^2 \left(\frac{\nu}{u_\infty \delta}\right)^{\frac{1}{4}}$		<u>Rohr</u> $\tau_0 = 0,0225 \rho u_{max}^2 \left(\frac{\nu}{u_{max} R}\right)^{\frac{1}{4}}$													

Fluidkinematik

Fluidbewegung

Langrange

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + u \frac{\partial f}{\partial x} + v \frac{\partial f}{\partial y} + w \frac{\partial f}{\partial z} = \frac{\partial f}{\partial t} + \underline{v} \cdot \nabla(f)$$

Euler

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}$$

Rohrhydraulik

Bernoulli-Gleichung

Stationär	Mittelwerte Stationär	Tatsächlich Instationär
$\frac{u_1^2}{2g} + \frac{p_1}{\gamma} + z_1 = \frac{u_2^2}{2g} + \frac{p_2}{\gamma} + z_2 + H_{E,1 \rightarrow 2}$ $\frac{\rho}{2} u_1^2 + p_1 + \gamma z_1 = \frac{\rho}{2} u_2^2 + p_2 + \gamma z_2 + \Delta p_{V,1 \rightarrow 2}$ $\Delta p_{V,1 \rightarrow 2} = \gamma H_{E,1 \rightarrow 2}$	$\bar{u} = \frac{1}{T} \int_{t_1}^{t_1+T} u(t) dt$ <u>Reynoldszahl</u> $Re_D = \frac{\bar{u} D}{\nu}$	$\frac{u_1^2}{2g} + \frac{p_1}{\gamma} + z_1 = \frac{u_2^2}{2g} + \frac{p_2}{\gamma} + z_2 + \frac{1}{g} \int_{s_1}^{s_2} \frac{\partial u}{\partial t} ds + H_{E,1 \rightarrow 2}$ <u>Universelles Rohrreibungsgesetz</u> $\Delta p_{V,1 \rightarrow 2} = \lambda \frac{L \rho}{2D} \bar{u}^2 \quad \left \quad h_{r,1 \rightarrow 2} = \lambda \frac{L \bar{u}^2}{2Dg}$

Stromröhre	Hydraulisch glattes Rohr ($\frac{k_s u_\tau}{\nu} \leq 5$)		
$\frac{\rho}{2} \alpha \bar{u}_1^2 + p_1 + \gamma z_1 = \frac{\rho}{2} \alpha \bar{u}_2^2 + p_2 + \gamma z_2 + \Delta p_{V,1 \rightarrow 2}$ $\bar{u} = \frac{1}{A} \int_A u dA = \frac{Q}{A} = \frac{\dot{V}}{A}$ Laminare Strömung: $\alpha = 2$ Turbulente Strömung: $\alpha \approx 1$	<u>Laminar</u> <u>Hagen-Poiseuille</u> $\lambda = \frac{64}{Re_D}$ $Re_{krit} = 2,3 \cdot 10^3 - 3 \cdot 10^3$	<u>Turbulent</u> <u>Prandtl-Beziehung</u> $\frac{1}{\sqrt{\lambda}} = 2 \log(Re \sqrt{\lambda}) - 0,8$ $Re > Re_{krit}$ Logarithmisches Geschwindigkeitsgesetz	<u>Blasius-Beziehung</u> $\lambda = 0,3164 Re^{\frac{1}{4}}$ $Re_{krit} < Re < 10^5$ $\frac{1}{7}\text{-Potenzgesetz}$

<u>Hydraulischer Durchmesser</u>		<u>Hydraulisch raues Rohr</u>																		
$D = d_h = 4 \frac{A}{U}$		<u>Sandrauhigkeiten</u>	<u>Übergangsbereich</u>	<u>Für $\frac{k_s u_\tau}{\nu} \leq 70$</u>																
<u>Querschnitt</u>	<u>Wert für d_h</u>	<table><tr><td><u>Oberfläche</u></td><td><u>k_s [mm]</u></td></tr><tr><td>Genieteter Stahl</td><td>0,9 - 9</td></tr><tr><td>Eisenbeton</td><td>0,3 - 3</td></tr><tr><td>Holz</td><td>0,2 - 0,9</td></tr><tr><td>Gusseisen</td><td>0,25</td></tr><tr><td>Verzinktes Eisen</td><td>0,15</td></tr><tr><td>Baustahl</td><td>0,05</td></tr><tr><td>Gezogene Rohre</td><td>0,0015</td></tr></table>	<u>Oberfläche</u>	<u>k_s [mm]</u>	Genieteter Stahl	0,9 - 9	Eisenbeton	0,3 - 3	Holz	0,2 - 0,9	Gusseisen	0,25	Verzinktes Eisen	0,15	Baustahl	0,05	Gezogene Rohre	0,0015	$\frac{1}{\sqrt{\lambda}} = 1,14 - 2 \log \left(\frac{k_s}{D} + \frac{9,35}{Re \sqrt{\lambda}} \right)$	$\lambda = \left(2 \log \left(\frac{R}{k_s} \right) + 1,74 \right)^{-2}$
<u>Oberfläche</u>	<u>k_s [mm]</u>																			
Genieteter Stahl	0,9 - 9																			
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Kreis	D		<u>Wandschubspannung</u>	<u>Für $Re \rightarrow \infty$</u>																
Quadrat	H																			
Dreieck	$\frac{S}{\sqrt{3}}$																			
Rechteck	$\frac{2 a b}{a + b}$																			
Kreisring	$d_a - d_i$																			
Spalt	$2 H$		$\tau_0 = \frac{\lambda}{8} \rho \bar{u}^2$	$\lambda = \left(1,14 - 2 \log \left(\frac{k_s}{D} \right) \right)^{-2}$																
Ellipse	$\begin{cases} 1,3 H & \left(\frac{B}{H} = 2 \right) \\ 1,57 H & \left(\frac{B}{H} \rightarrow \infty \right) \end{cases}$		<u>Rohrleitungsverluste</u>																	
		$\Delta p_{V, \nu} = \zeta_\nu \frac{\rho}{2} \bar{u}_\nu^2$	$h_\nu = \zeta_\nu \frac{\bar{u}_\nu^2}{2 g}$	$\bar{u}_n u = \frac{Q}{A_\nu}$																

Temperaturabhängigkeit der Viskosität	Leistung von Turbine und Pumpe		
$\frac{\eta}{\eta_0} = \exp\left(\frac{T_A}{T_A + T_B} - \frac{T_A}{T_0 + T_B}\right)$ $T_0 = 273,15 \text{ K}$ Wasser: $T_A = 506 \text{ K}$ $T_B = -150 \text{ K}$	<u>Allgemein</u> $P = \Delta p A \bar{u} = \Delta p Q = \gamma Q h$ $Q = \dot{V}$ $\gamma = \rho g$	<u>Mit Wirkungsgrad</u> <u>Turbine</u> $(\Delta p)_T = \frac{P_T}{\eta_T Q}$ $h_T = \frac{P_T}{\eta_T \gamma Q}$	<u>Pumpe</u> $(\Delta p)_P = -\frac{\eta_P P_P}{Q}$ $h_P = -\frac{\eta_P P_P}{\gamma Q}$

Verluste durch Rohrgeometrie

Querschnittsänderung

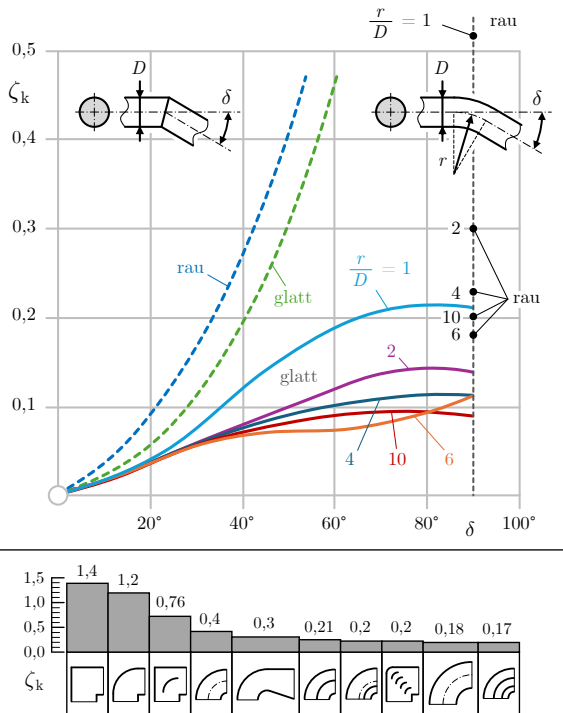
Eintritt		
Eintrittskante	ζ_e	
a) Scharf	0,5	
Gebrochen	0,25	
Gut Abgerundet	0,06	
b) Vorspringend	< 3,0	
Austritt		
Turbulent: $\zeta_a \approx 1,0$		
Laminar: $\zeta_a \approx 2,0$		

		Schieber							
$\frac{s}{D}$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{3}{8}$	$\frac{4}{8}$	$\frac{5}{8}$	$\frac{6}{8}$	$\frac{7}{8}$	$\frac{8}{8}$	
$\frac{A_s}{A}$	0,948	0,856	0,743	0,609	0,466	0,315	0,159	0	
ζ_s	0,07	0,26	0,81	2,06	5,52	17,0	97,8		

Rohrhydraulik

Verluste durch Rohrgeometrie

Krümmen



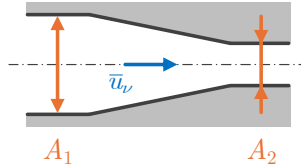
Verengung

Allmähliche Verengung (Düse)

$$\Delta p_{\text{Düse}} = \frac{\rho}{2} \bar{u}_2^2 \left[1 - \left(\frac{A_2}{A_1} \right)^2 \right]$$

Druckverlustbeiwert

$$\zeta_f \approx 0$$

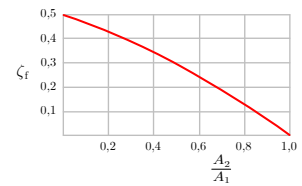
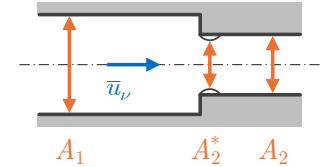


Plötzliche Verengung

$$\Delta p_{\text{Eng}} = \frac{\rho}{2} \bar{u}_2^2 \left[1 - \left(\frac{A_2^*}{A_1} \right)^2 \right]$$

Druckverlustbeiwert

$$\zeta_f = \left(\frac{A_2}{A_2^*} - 1 \right)^2 = \left(\frac{1}{\mu} - 1 \right)^2$$



Diffusor

Allmähliche Erweiterung

Ideal

$$\Delta p_{\text{ideal}} = \frac{\rho}{2} \bar{u}_1^2 \left[1 - \left(\frac{A_1}{A_2} \right)^2 \right]$$

Real

$$\Delta p_{\text{real}} = \eta \frac{\rho}{2} \bar{u}_1^2 \left[1 - \left(\frac{A_1}{A_2} \right)^2 \right]$$

Diffusorwirkungsgrad

$$\eta = \frac{\Delta p_{\text{real}}}{\Delta p_{\text{ideal}}}$$

Druckverlustbeiwert

$$\zeta = (0,12 \dots 0,2) \left[1 - \left(\frac{A_1}{A_2} \right)^2 \right]$$

Plötzliche Erweiterung

Ideal

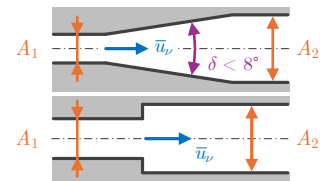
$$\Delta p_{\text{ideal}} = \frac{\rho}{2} \bar{u}_1^2 \left[1 - \left(\frac{A_1}{A_2} \right)^2 \right]$$

Real

$$\Delta p_{\text{Carnot}} = \frac{\rho}{2} \bar{u}_1^2 \left[2 \frac{A_1}{A_2} \left(1 - \frac{A_1}{A_2} \right) \right]$$

Druckverlustbeiwert

$$\zeta = \left(1 - \frac{A_1}{A_2} \right)^2$$



Moody-Diagramm

<u>Laminar</u> ① $\lambda = \frac{64}{\text{Re}}$	<u>Blasius</u> ② $\lambda = \frac{0,3164}{\sqrt[4]{\text{Re}}}$	<u>Prandtl</u> ② $\frac{3}{\sqrt{\lambda}} = 2 \log_{10} \left(\text{Re} \sqrt{\lambda} \right) - 0,8$	<u>Übergangsbereich</u> ④ $\frac{1}{\sqrt{\lambda}} = 1,14 - 2 \log_{10} \left(\frac{k_S}{D} + \frac{9,35}{\text{Re} \sqrt{\lambda}} \right)$	<u>Turbulent & Rau</u> ⑤ $\lambda = \left(1,14 - 2 \log_{10} \left(\frac{k_S}{D} \right) \right)^{-2}$
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